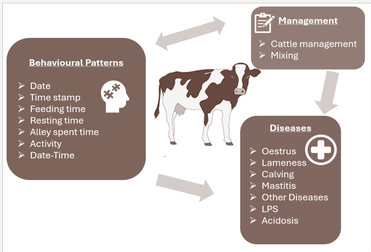
Investigations on Multi Disease Prediction Framework for Dairy Cows Using Time Series Behavioural Analytics

Vishnu Bhuvanachandran Nair Rajeswari Amma
P G Scholar, M. Sc. Data Analytics, Department of Mathematics and Computing Sciences, Dundalk Institute of Technology

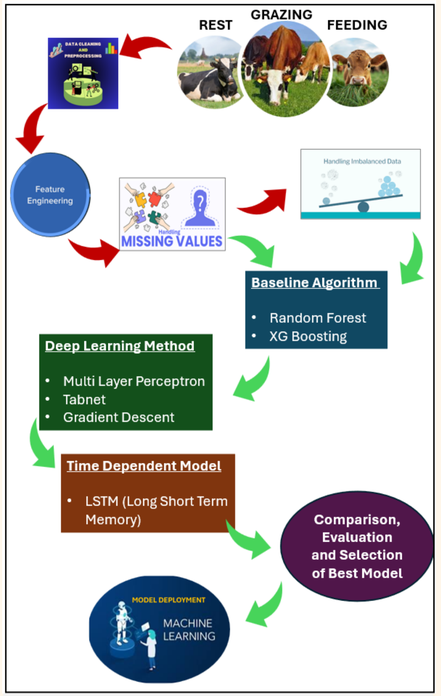
Introduction

Development of agricultural sector is an integral part of every nations economical development which directly impacts GDP. Apart from the farming and crop development, livestock farming forms the inevitable part of agricultural economies in European Union, of which the 1.4% of total GDP was contributed by agriculture. The timely disease diagnosis via. conventional methods such as veterinarian intervention has many challenges such as (1) availability of medical facilities, (2) remote location of farms, and (3) awareness of farmers. Therefore this work aims to address these challenges by developing an data driven machine learning algorithm for disease prediction using behavioural and activity-level data collected from cattle. This prediction is carried out by multi-output classification, which allows simultaneous prediction of multiple diseases the same time with respect to the behavioural data of the cattle.



Cattle behaviour and diseases from the dataset

Objectives



Step by Step Objectives of the Project

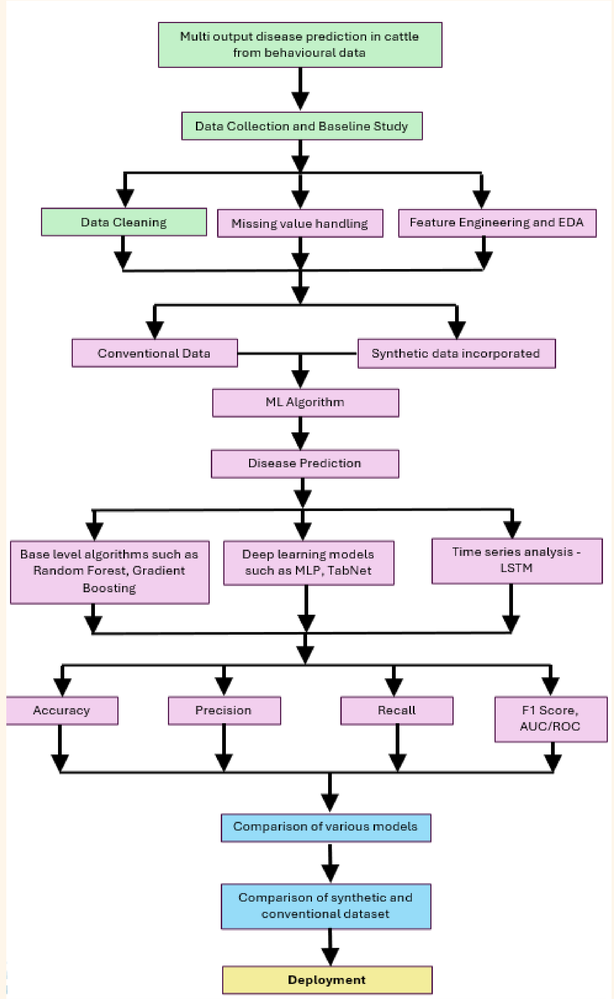
Dataset

INRAE
science for people, life & earth

The dataset for the project was selected from the publicly available French agricultural database.

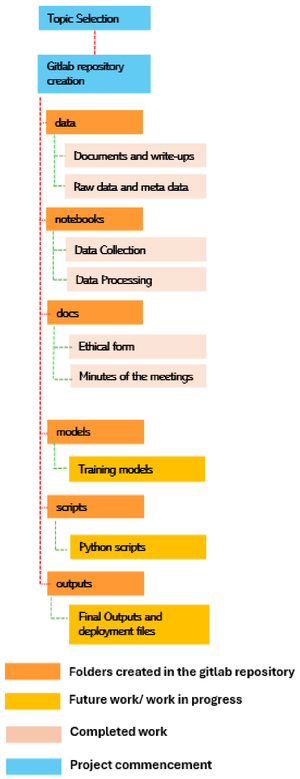
https://entrepot.recherche.data.gouv.fr/dataset.xhtml?persistentId=doi:10.15454/52J8YS

Methodology



Workflow for the Methodology of the Project

Artefact



Data Imbalance

Disease Condition	Absent	Present
Oestrus	2,343,401	7,944
Calving	2,347,049	4,296
Lameness	2,348,129	3,216
Mastitis	2,350,289	1,056
Other Disease	2,349,329	2,016
Accidents	2,350,985	360
Disturbance	2,037,783	313,562
Mixing	2,349,617	1,728

Count of disease present/ absent conditions

Strengths

- Larger dataset (300+ cows) with behavioral data with time characteristics
- Multi-output prediction, enabling simultaneous detection of multiple health events at the same time.
- Data imbalance handling via. Oversampling or undersampling
- Comparison between traditional models and deep learning models such as MLP (Multi Layer Perceptron) and TabNet
- Analysis of the effectiveness of the time dependence via. the neural methods like LSTM (Long Short Term Memory)

References

[1] Rasmussen, P., Barkema, H.W., Osei, P.P., Taylor, J., Shaw, A.P., Conrady, B., Chaters, G., Muñoz, V., Hall, D.C., Apenteng, O.O. and Rushton, J., 2024. Global losses due to dairy cattle diseases: A comorbidity-adjusted economic analysis. Journal of Dairy Science, 107(9), pp.6945-6970.

[2] Giannakis, E. and Bruggeman, A., 2015. The highly variable economic performance of European agriculture. Land Use Policy, 45, pp.26-35.

[3] Holloway, L., Mahon, N., Clark, B. and Proctor, A., 2023. Changing interventions in farm animal health and welfare: A governmentality approach to the case of lameness. Journal of Rural Studies, 97, pp.95-104.

[4] Berrada, M., Raboisson, D. and Lhermie, G., 2024. Determinants of access to animal health care in France: evidence from a spatial econometric framework. Review of Agricultural, Food and Environmental Studies, pp.1-25.

[5] Pavkin DY, Dorokhov AS, Vladimirov FE, Dovlatov IM, Lyalin KS. Algorithms for detecting cattle diseases at early stages and for making diagnoses and related recommendations. Applied Sciences. 2021 Nov 24;11(23):11148.